

Sophia Freire Aparicio Oliveira - 133141 e Lucas Alves de Lima - 129199

ex1:



The screenshot shows a web application interface with a dark theme. At the top, there are two tabs: "Playground Result: test.usuarios" and "Playground Result: streamflix.usuarios". Below the tabs, there is a toolbar with buttons for "Insert Document", "Refresh", and pagination controls showing "10" items, "1-10" range, and navigation arrows. The main content area displays three JSON objects, each representing a user. Each object is expanded, showing its properties: _id (ObjectId), nome (name), email, idade (age as NumberInt), cidade (city), estado (state), interesses (interests), ativo (active), and endereco (address). The users listed are Ana Souza, Carlos Lima, and Fernanda Rocha.

```
▼ {
  _id: ObjectId('6a3e8b08ab9d566c0e0927db'),
  nome: 'Ana Souza',
  email: 'ana@email.com',
  idade: NumberInt('22'),
  cidade: 'Curitiba',
  estado: 'PR',
  > interesses: [ ...
    ],
  ativo: true,
  > endereco: { ...
    }
}
```

```
▼ {
  _id: ObjectId('6a3e8b08ab9d566c0e0927dc'),
  nome: 'Carlos Lima',
  email: 'carlos@email.com',
  idade: NumberInt('31'),
  cidade: 'Maringá',
  estado: 'PR',
  > interesses: [ ...
    ],
  ativo: false,
  > endereco: { ...
    }
}
```

```
▼ {
  _id: ObjectId('6a3e8b08ab9d566c0e0927dd'),
  nome: 'Fernanda Rocha',
  email: 'fernanda@email.com',
  idade: NumberInt('27'),
  cidade: 'Londrina',
  estado: 'PR',
  > interesses: [ ...
    ],
  ativo: true
}
```

The bottom of the image shows a Windows taskbar with various application icons (Spotify, Chrome, VS Code) and system tray information including the location "POR PTB2", signal strength, volume, battery level at 90%, and the date/time "09:52 30/06/2026".

```

{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce49'),
  titulo: 'Interestelar',
  tipo: 'filme',
  ano: NumberInt('2014'),
  generos: [ ... ],
  avaliacaoMedia: Double('9.5'),
  duracaoMinutos: NumberInt('169'),
  disponivel: true,
  visualizacoes: NumberInt('2500000'),
  diretor: { ... },
  elenco: [ ... ]
}

{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce4a'),
  titulo: 'Matrix',
  tipo: 'filme',
  ano: NumberInt('1999'),
  generos: [ ... ],
  avaliacaoMedia: Double('9.2'),
  duracaoMinutos: NumberInt('136'),
  disponivel: true,
  visualizacoes: NumberInt('3100000'),
  diretor: { ... },
  elenco: [ ... ]
}

{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce4b'),
  titulo: 'Avatar',
  tipo: 'filme',
  ano: NumberInt('2009')
}
```

```
C:\Users> sophy > JS playground-2.mongodb.js > ...
Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use('streamflix')
14 db.usuarios.find({ cidade: "curitiba" })
15
```

```
+ Insert Document Refresh 10 1-3 < >
{
  _id: ObjectId('6a3e8b08ab9d566c0e0927db'),
  nome: 'Ana Souza',
  email: 'ana@email.com',
  idade: NumberInt('22'),
  cidade: 'Curitiba',
  estado: 'PR',
  interesses: [...],
  ativo: true,
  endereco: {...}
}

{
  _id: ObjectId('6a3e8b08ab9d566c0e0927de'),
  nome: 'João Mendes',
  email: 'joao@email.com',
  idade: NumberInt('19'),
  cidade: 'Curitiba',
  estado: 'PR',
  interesses: [...],
  ativo: true
}

{
  _id: ObjectId('6a3e8b08ab9d566c0e0927e4'),
  nome: 'Eduardo Martins',
  email: 'eduardo@email.com',
  idade: NumberInt('33'),
  cidade: 'Curitiba',
  estado: 'PR',
  interesses: [...],
  ativo: true,
  premium: false
}
```

ex4:

```
Welcome JS playground-2.mongodb.js ... Playground Result: streamflix.usuarios Playground Result: streamflix conteudos X
C:\Users> sophy > JS playground-2.mongodb.js > ...
Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use('streamflix')
14 db.conteudos.find({ tipo: "filme" })
```

```
+ Insert Document Refresh 10 1-10 < >
{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce49'),
  titulo: 'Interestelar',
  tipo: 'filme',
  ano: NumberInt('2014'),
  generos: [...],
  avaliacaoMedia: Double('9.5'),
  duracaoMinutos: NumberInt('169'),
  disponivel: true,
  visualizacoes: NumberInt('2500000'),
  diretor: {...},
  elenco: [...],
}

{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce4a'),
  titulo: 'Matrix',
  tipo: 'filme',
  ano: NumberInt('1999'),
  generos: [...],
  avaliacaoMedia: Double('9.2'),
  duracaoMinutos: NumberInt('136'),
  disponivel: true,
  visualizacoes: NumberInt('3100000'),
  diretor: {...},
  elenco: [...],
}

{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce4b'),
  titulo: 'Avatar',
  tipo: 'filme',
  ano: NumberInt('2009')
}
```

ex5:

```
JS playground-2.mongodb.js X
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to be able
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with a limit of
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.findOne({ titulo: "Matrix" })

1 {
2   _id: ObjectId('6a3e87ce3c1b96d4f3acce4a'),
3   titulo: 'Matrix',
4   tipo: 'filme',
5   ano: NumberInt('1999'),
6   generos: [
7     'Ação',
8     'Ficção Científica'
9   ],
10  avaliacaoMedia: Double('9.2'),
11  duracaoMinutos: NumberInt('136'),
12  disponivel: true,
13  visualizacoes: NumberInt('3100000'),
14  diretor: {
15    nome: 'Lana Wachowski e Lilly Wachowski',
16    pais: 'Estados Unidos'
17  },
18  elenco: [
19    'Keanu Reeves',
20    'Laurence Fishburne',
21    'Carrie-Anne Moss'
22  ]
23 }
```

ex6:

```
JS playground-2.mongodb.js X
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to be able
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with a limit of
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.usuarios.insertOne({
15   nome: "Seu Nome",
16   email: "seuemail@email.com",
17   idade: 25,
18   cidade: "Maringá",
19   estado: "PR",
20   interesses: ["ação", "comédia"],
21   ativo: true
22 })

1 {
2   acknowledged: true,
3   insertedId: ObjectId('6a43bf19217026f088119448')
4 }
```

ex7:

The screenshot shows a VS Code editor with a file named `playground-2.mongodb.js`. The script is a MongoDB playground template. It connects to a cluster, selects the `streamflix` database, and inserts a document into the `conteudos` collection. The document has the following fields: `titulo` (title), `tipo` (type), `ano` (year), `generos` (genres), `avaliacaoMedia` (average rating), `duracaoMinutos` (duration in minutes), and `disponivel` (available).

```
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB Playground
4 // Make sure you are connected to enable completions and to get MongoDB
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right side
7 // By default the first 20 documents will be returned with the sort order
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.insertOne({
15   titulo: "O Senhor dos Anéis",
16   tipo: "filme",
17   ano: 2001,
18   generos: ["fantasia", "aventura"],
19   avaliacaoMedia: 8.8,
20   duracaoMinutos: 178,
21   disponivel: true
22 })
```

The right side of the editor shows the result of the `insertOne` command. It is a JSON object with the following fields: `acknowledged` (true), `insertedId` (ObjectId('6a43bf5757fd64b277172596')).

```
1 {
2   acknowledged: true,
3   insertedId: ObjectId('6a43bf5757fd64b277172596')
4 }
```

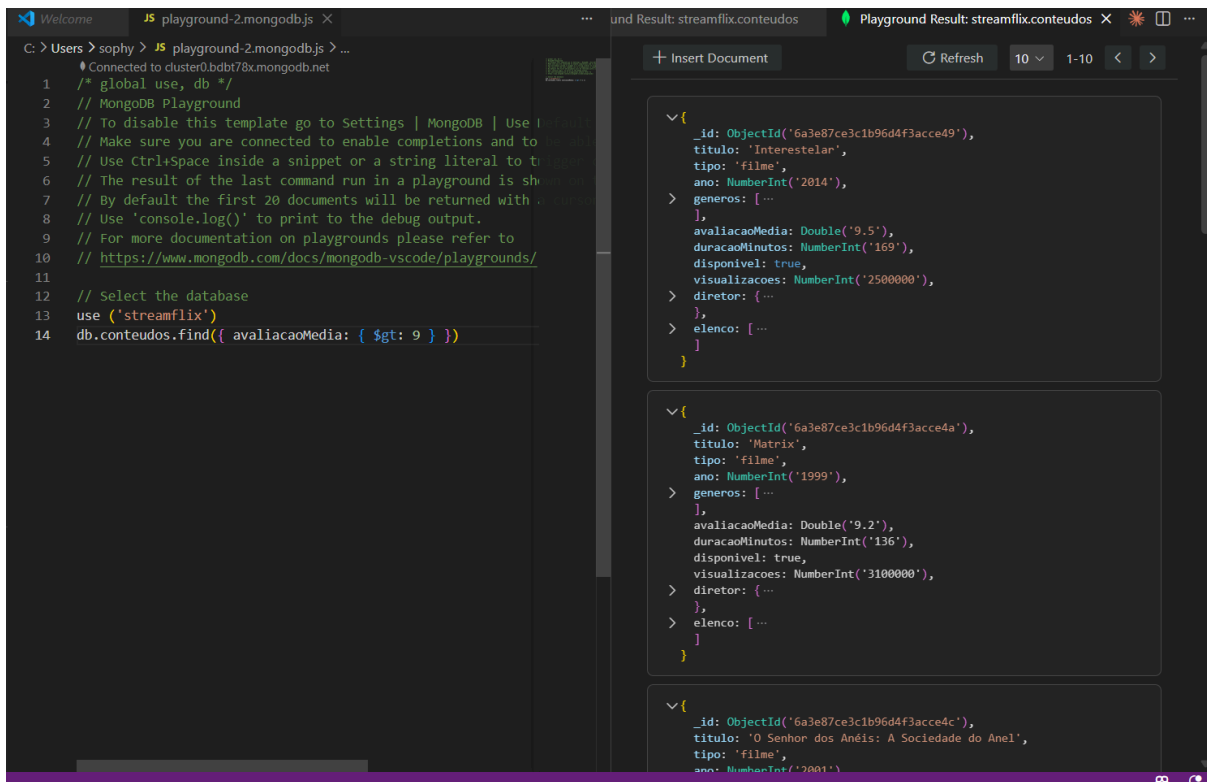
The screenshot shows a VS Code editor with a file named `playground-2.mongodb.js`. The script is a MongoDB playground template. It connects to a cluster, selects the `streamflix` database, and finds documents in the `conteudos` collection where the `titulo` is "O Senhor dos Anéis".

```
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB Playground
4 // Make sure you are connected to enable completions and to get MongoDB
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right side
7 // By default the first 20 documents will be returned with the sort order
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.find({ titulo: "O Senhor dos Anéis" })
```

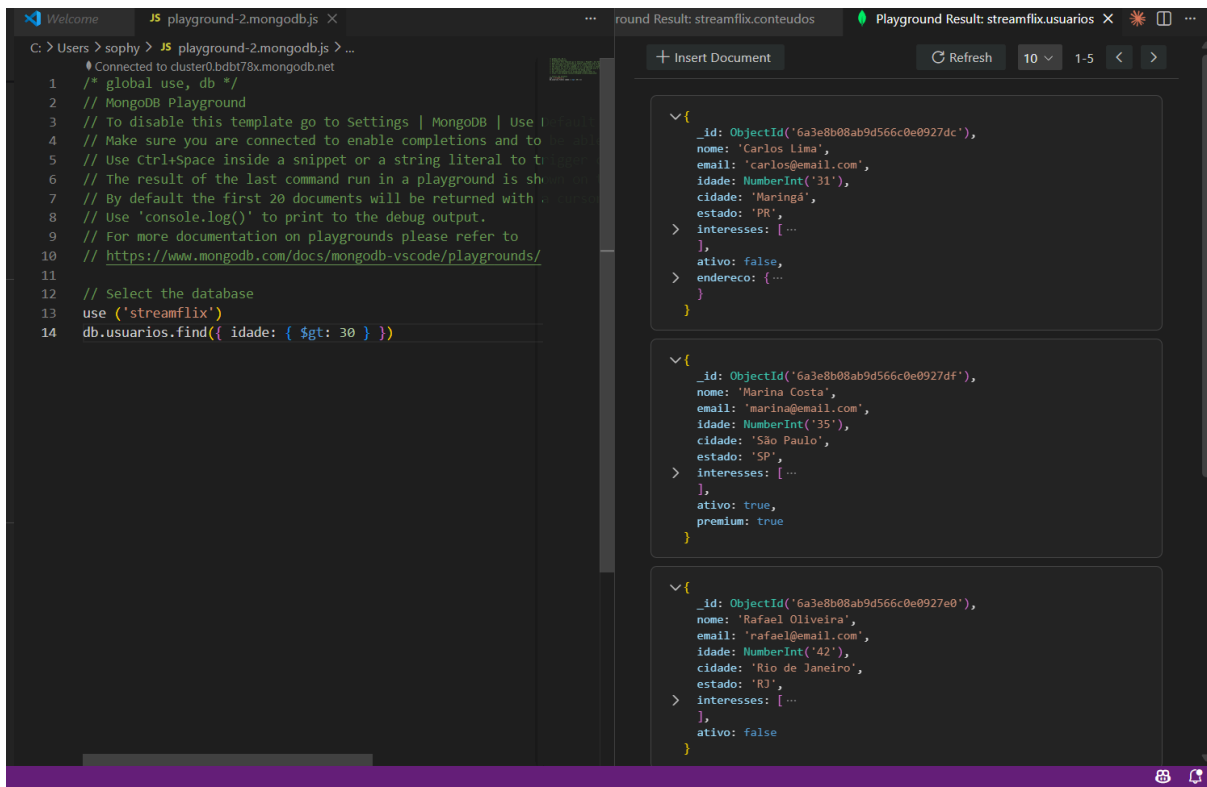
The right side of the editor shows the result of the `find` command. It is a JSON array containing one document with the following fields: `_id` (ObjectId), `titulo` (title), `tipo` (type), `ano` (year), `generos` (genres), `avaliacaoMedia` (average rating), `duracaoMinutos` (duration in minutes), and `disponivel` (available).

```
1 {
2   _id: ObjectId('6a43bf5757fd64b277172596'),
3   titulo: 'O Senhor dos Anéis',
4   tipo: 'filme',
5   ano: NumberInt('2001'),
6   generos: [ ... ],
7   avaliacaoMedia: Double('8.8'),
8   duracaoMinutos: NumberInt('178'),
9   disponivel: true
10 }
```

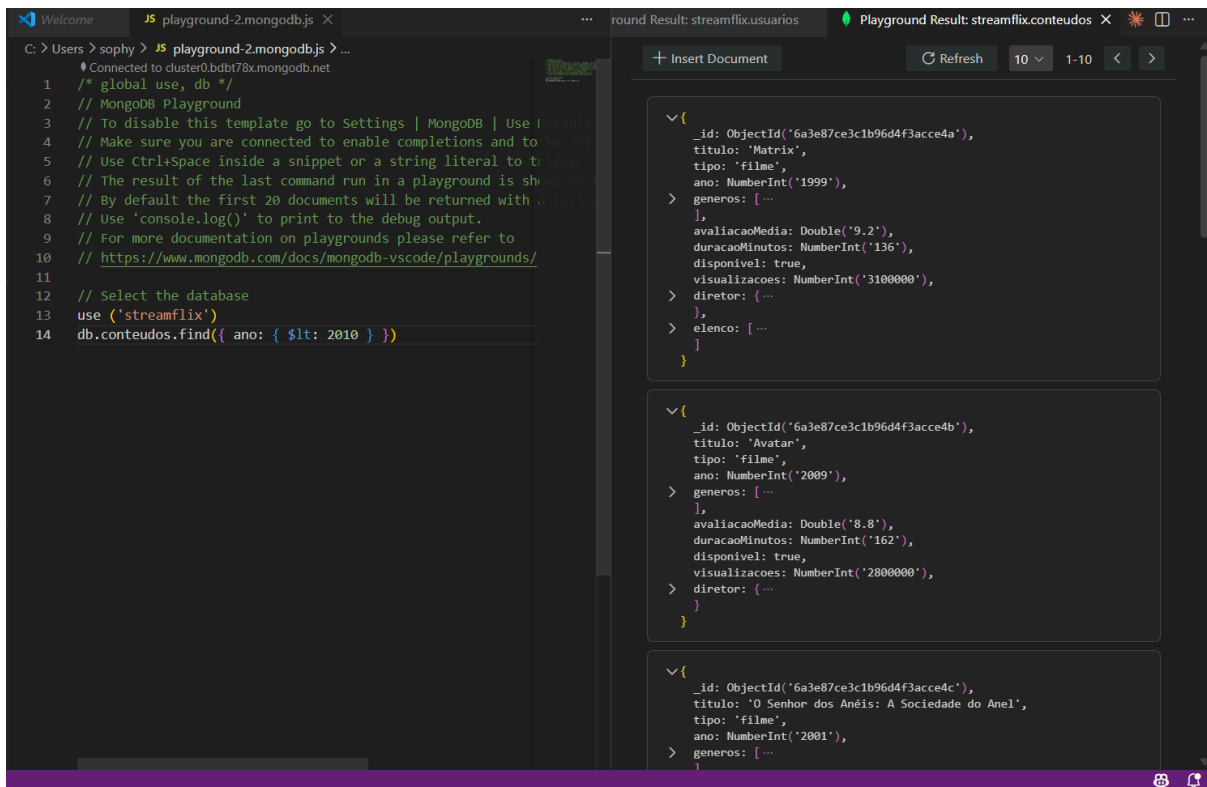
ex8:



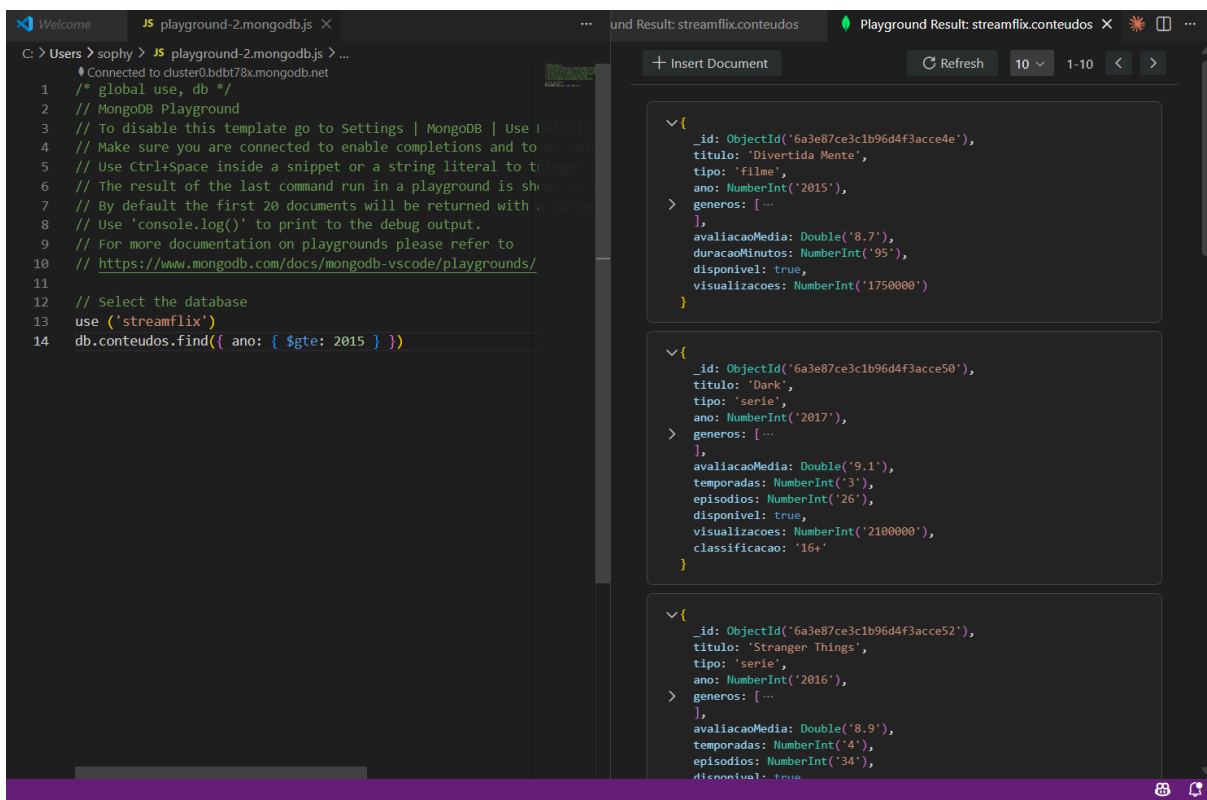
ex9:



ex10:



ex11:



ex12:

```

C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1  /* global use, db */
2  // MongoDB Playground
3  // To disable this template go to Settings | MongoDB | Use Default
4  // Make sure you are connected to enable completions and to the db
5  // Use Ctrl+Space inside a snippet or a string literal to trigger
6  // The result of the last command run in a playground is shown on
7  // By default the first 20 documents will be returned with a cursor
8  // Use 'console.log()' to print to the debug output.
9  // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.find({ avaliacaoMedia: { $lte: 8.8 } })

```

```

+ Insert Document
Refresh 10 1-10 < >
[
  {
    _id: ObjectId('6a3e87ce3c1b96d4f3acce4b'),
    titulo: 'Avatar',
    tipo: 'filme',
    ano: NumberInt('2009'),
    generos: [ ... ],
    avaliacaoMedia: Double('8.8'),
    duracaoMinutos: NumberInt('162'),
    disponivel: true,
    visualizacoes: NumberInt('2800000'),
    diretor: { ... }
  },
  {
    _id: ObjectId('6a3e87ce3c1b96d4f3acce4e'),
    titulo: 'Divertida Mente',
    tipo: 'filme',
    ano: NumberInt('2015'),
    generos: [ ... ],
    avaliacaoMedia: Double('8.7'),
    duracaoMinutos: NumberInt('95'),
    disponivel: true,
    visualizacoes: NumberInt('1750000')
  },
  {
    _id: ObjectId('6a3e87ce3c1b96d4f3acce53'),
    titulo: 'The Office',
    tipo: 'serie',
    ano: NumberInt('2005'),
    generos: [ ... ],
    avaliacaoMedia: Double('8.8'),
    temporadas: NumberInt('9'),
    episodios: NumberInt('201'),
    disponivel: true
  },
  ...
]

```

ex13:

```

C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1  /* global use, db */
2  // MongoDB Playground
3  // To disable this template go to Settings | MongoDB | Use Default
4  // Make sure you are connected to enable completions and to the db
5  // Use Ctrl+Space inside a snippet or a string literal to trigger
6  // The result of the last command run in a playground is shown on
7  // By default the first 20 documents will be returned with a cursor
8  // Use 'console.log()' to print to the debug output.
9  // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.usuarios.find({ estado: { $ne: "PR" } })

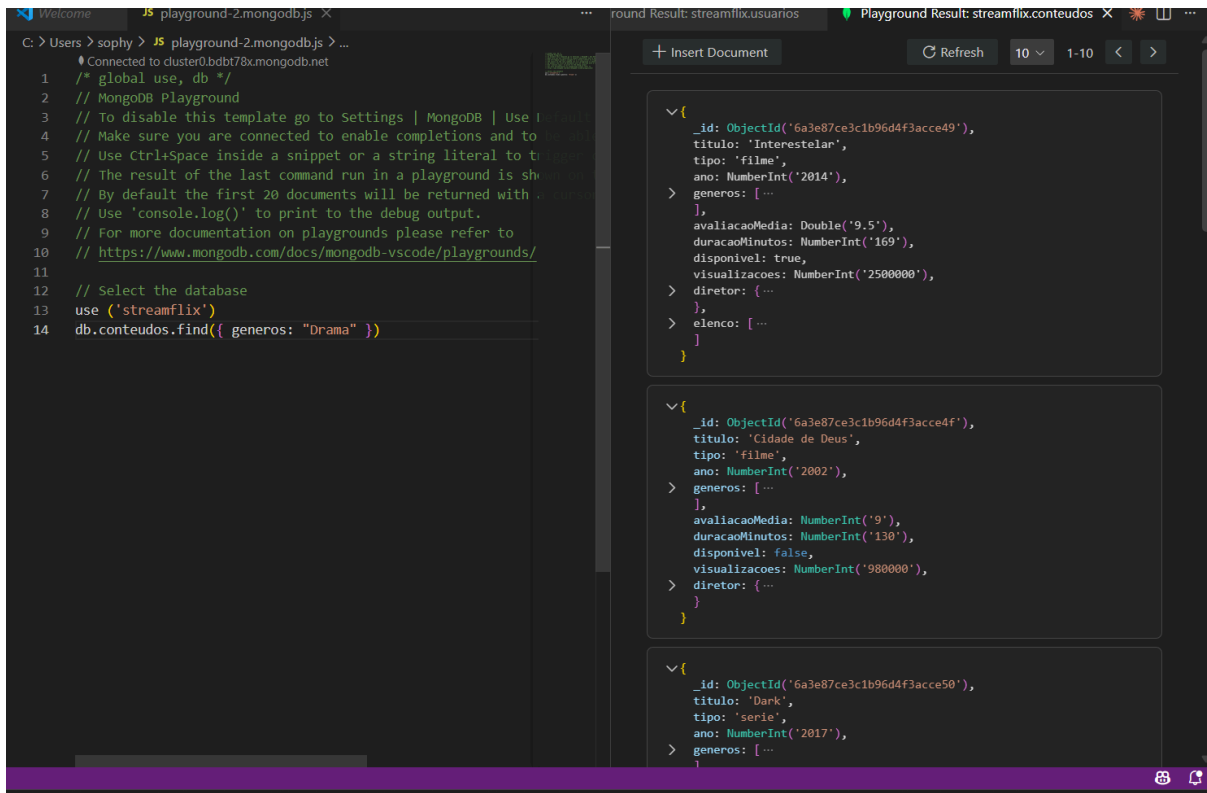
```

```

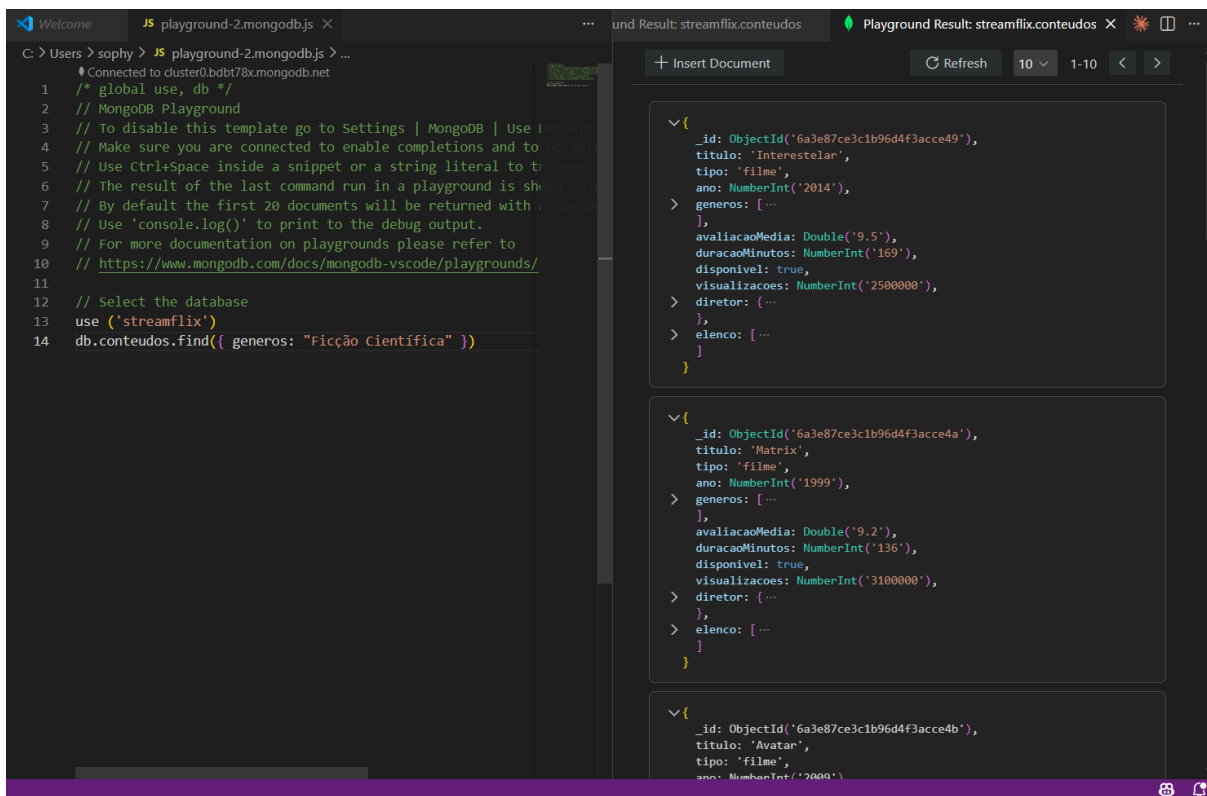
+ Insert Document
Refresh 10 1-4 < >
[
  {
    _id: ObjectId('6a3e8b08ab9d566c0e0927df'),
    nome: 'Marina Costa',
    email: 'marina@email.com',
    idade: NumberInt('35'),
    cidade: 'São Paulo',
    estado: 'SP',
    interesses: [ ... ],
    ativo: true,
    premium: true
  },
  {
    _id: ObjectId('6a3e8b08ab9d566c0e0927e0'),
    nome: 'Rafael Oliveira',
    email: 'rafael@email.com',
    idade: NumberInt('42'),
    cidade: 'Rio de Janeiro',
    estado: 'RJ',
    interesses: [ ... ],
    ativo: false
  },
  {
    _id: ObjectId('6a3e8b08ab9d566c0e0927e2'),
    nome: 'Lucas Ferreira',
    email: 'lucas@email.com',
    idade: NumberInt('29'),
    cidade: 'Florianópolis',
    estado: 'SC',
    interesses: [ ... ],
    ativo: true
  },
  ...
]

```

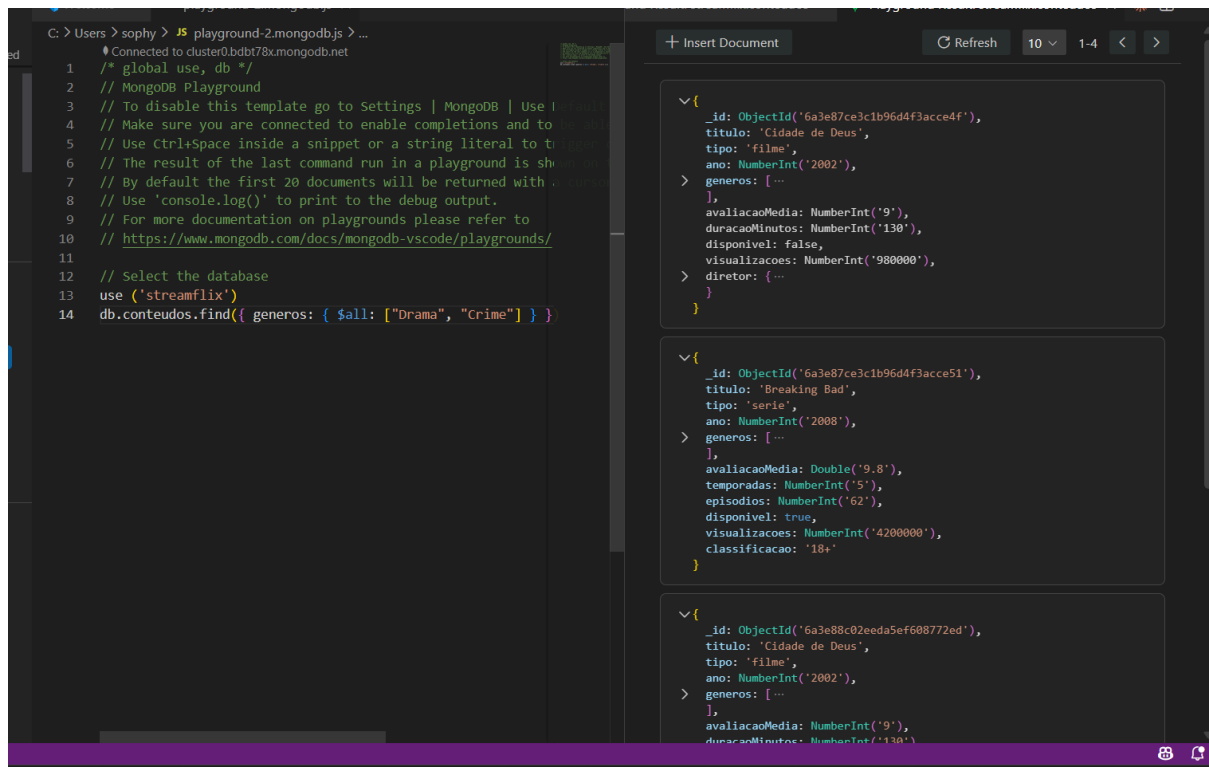
ex14:



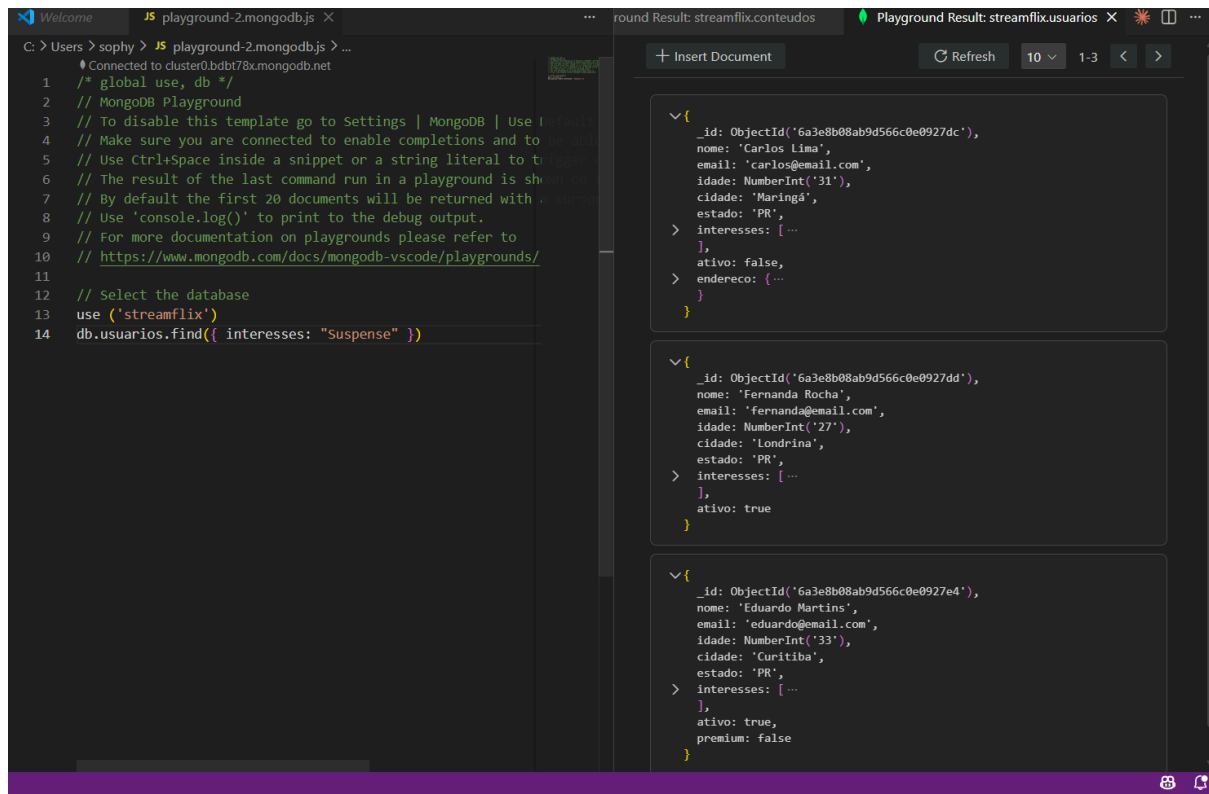
ex15:



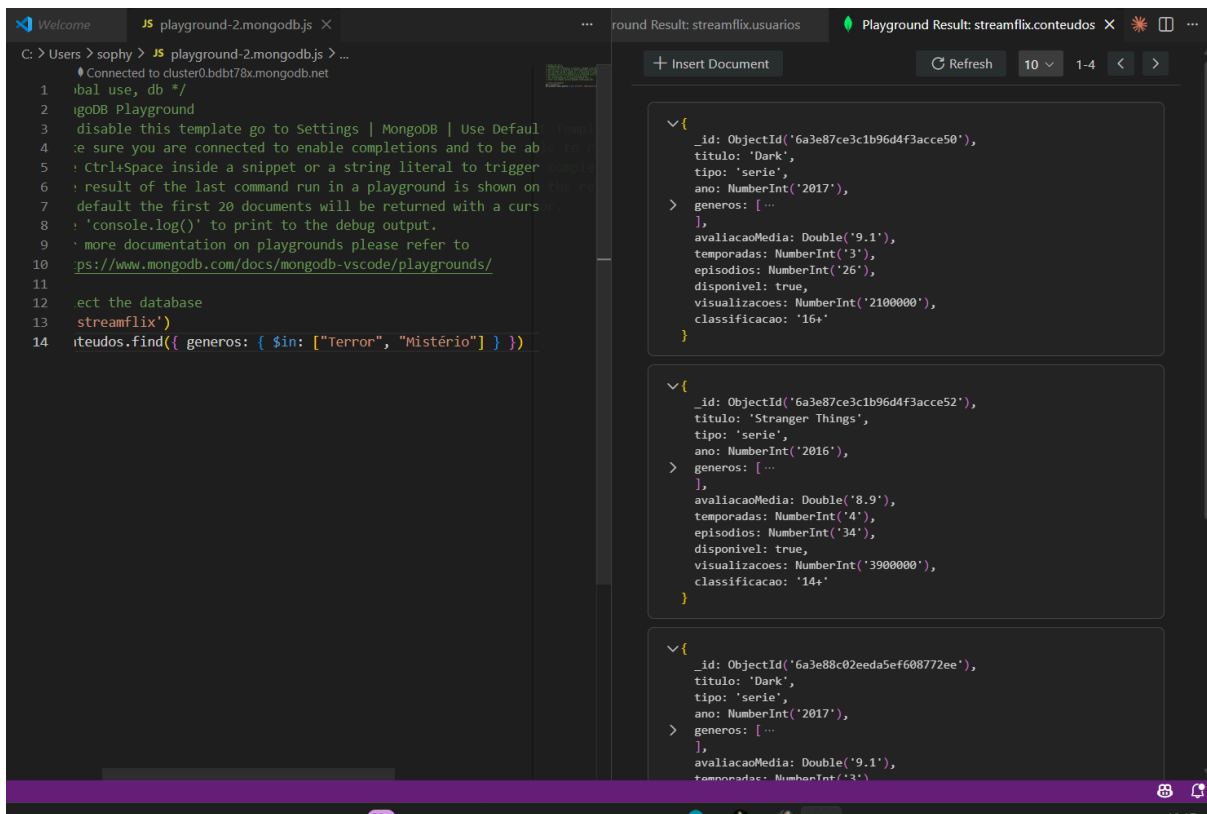
ex16:



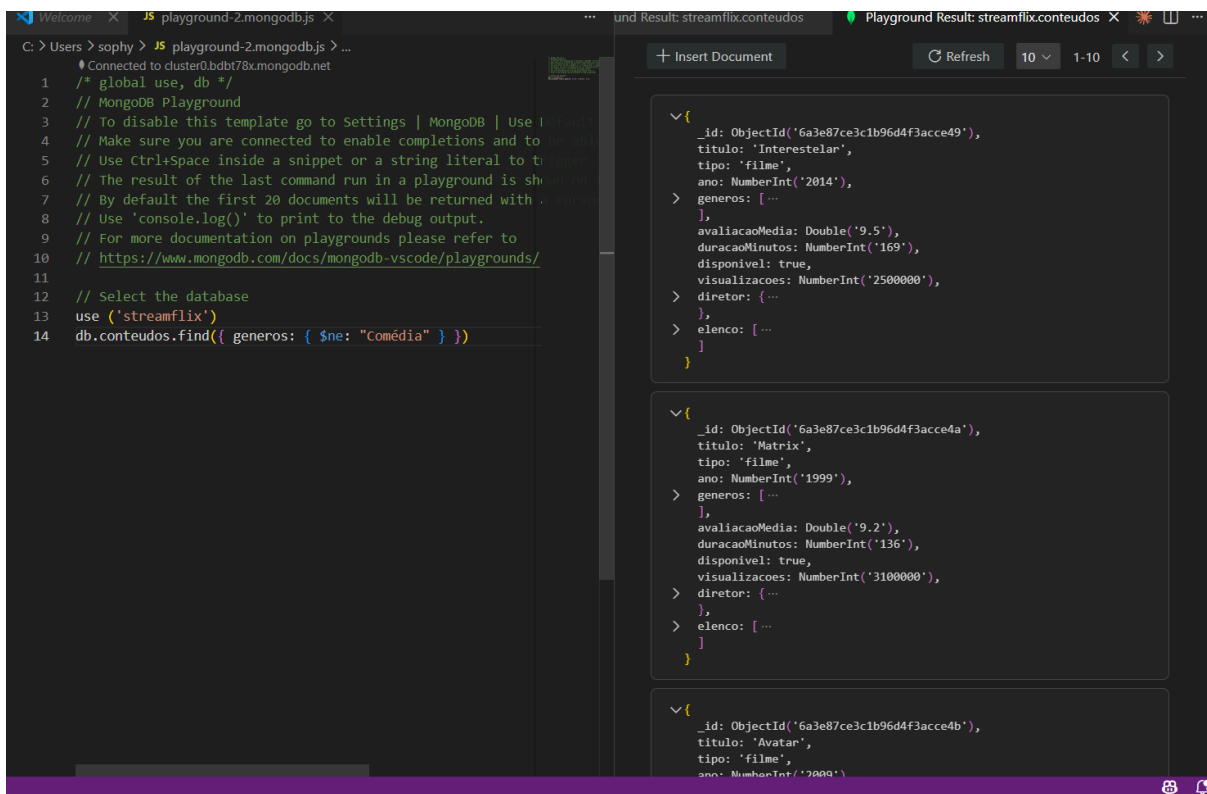
ex17:



ex18:



ex19:



ex20:

```

Welcome JS playground-2.mongodb.js x und Result: streamflix conteudos Playground Result: streamflix conteudos x
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use Default
4 // Make sure you are connected to enable completions and to get
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with a
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.find({ "diretor.nome": "Christopher Nolan" })

```

```

+ Insert Document Refresh 10 1-4 < >
{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce49'),
  titulo: 'Interstellar',
  tipo: 'filme',
  ano: NumberInt('2014'),
  > generos: [...],
  avaliacaoMedia: Double('9.5'),
  duracaoMinutos: NumberInt('169'),
  disponivel: true,
  visualizacoes: NumberInt('2500000'),
  > diretor: {...},
  > elenco: [...],
  }
}

{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce4d'),
  titulo: 'A Origem',
  tipo: 'filme',
  ano: NumberInt('2010'),
  > generos: [...],
  avaliacaoMedia: Double('9.1'),
  duracaoMinutos: NumberInt('148'),
  disponivel: true,
  visualizacoes: NumberInt('1950000'),
  > diretor: {...},
  }
}

{
  _id: ObjectId('6a3e88c02eeda5ef608772e7'),
  titulo: 'Interstellar',
  tipo: 'filme',
  ano: NumberInt('2014'),
  > generos: [...],
  }
}

```

ex21:

```

C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use Default
4 // Make sure you are connected to enable completions and to get
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with a
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.find({ "diretor.pais": "Reino Unido" })

```

```

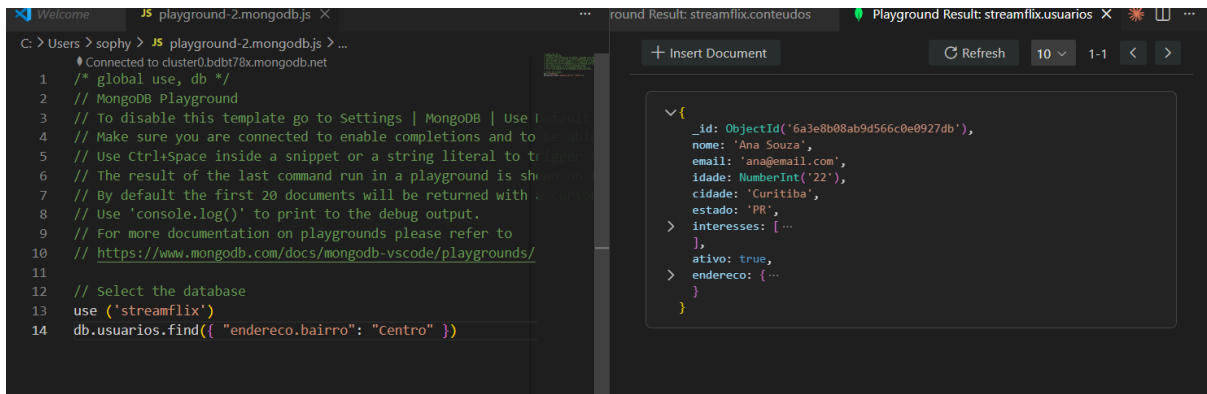
+ Insert Document Refresh 10 1-4 < >
{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce49'),
  titulo: 'Interstellar',
  tipo: 'filme',
  ano: NumberInt('2014'),
  > generos: [...],
  avaliacaoMedia: Double('9.5'),
  duracaoMinutos: NumberInt('169'),
  disponivel: true,
  visualizacoes: NumberInt('2500000'),
  > diretor: {...},
  > elenco: [...],
  }
}

{
  _id: ObjectId('6a3e87ce3c1b96d4f3acce4d'),
  titulo: 'A Origem',
  tipo: 'filme',
  ano: NumberInt('2010'),
  > generos: [...],
  avaliacaoMedia: Double('9.1'),
  duracaoMinutos: NumberInt('148'),
  disponivel: true,
  visualizacoes: NumberInt('1950000'),
  > diretor: {...},
  }
}

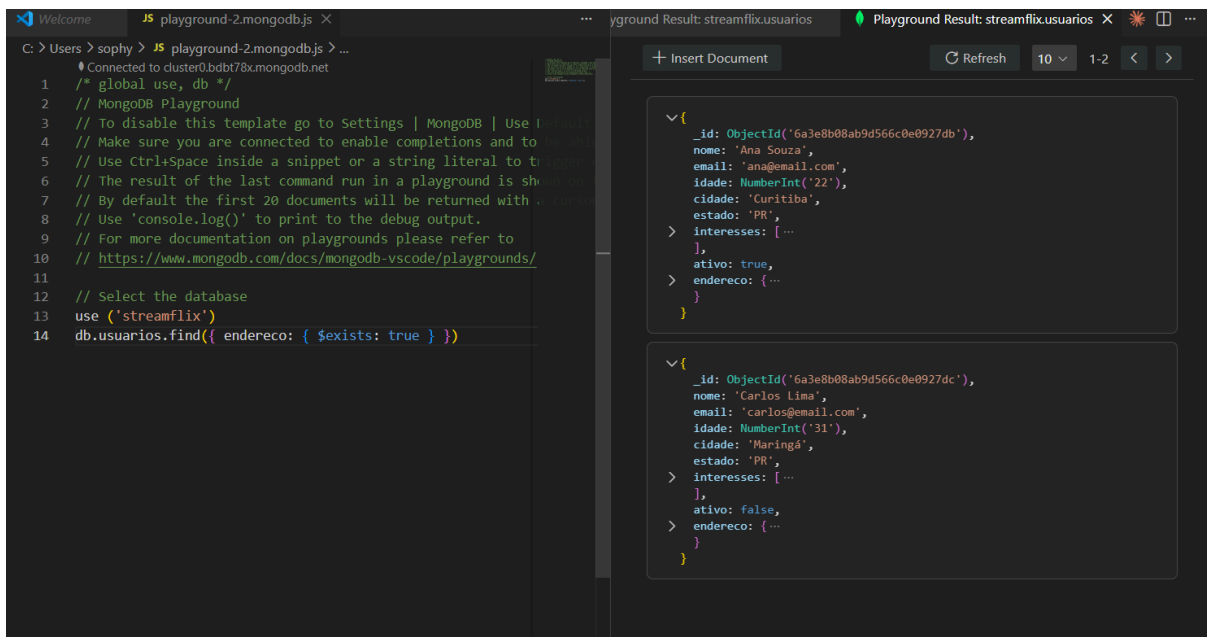
{
  _id: ObjectId('6a3e88c02eeda5ef608772e7'),
  titulo: 'Interstellar',
  tipo: 'filme',
  ano: NumberInt('2014'),
  > generos: [...],
  }
}

```

ex22:



ex23:



ex24:

The screenshot shows the MongoDB Playground interface. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and querying a collection. The right pane displays the results of the query, showing three documents from the 'streamflix.usuarios' collection.

```
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use Defaults
4 // Make sure you are connected to enable completions and to be able
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with a limit of
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.usuarios.find({ endereco: { $exists: false } })
```

Playground Result: streamflix.usuarios

```
{
  _id: ObjectId('6a3e8b08ab9d566c0e0927dd'),
  nome: 'Fernanda Rocha',
  email: 'fernanda@email.com',
  idade: NumberInt('27'),
  cidade: 'Londrina',
  estado: 'PR',
  interesses: [...],
  ativo: true
}

{
  _id: ObjectId('6a3e8b08ab9d566c0e0927de'),
  nome: 'João Mendes',
  email: 'joao@email.com',
  idade: NumberInt('19'),
  cidade: 'Curitiba',
  estado: 'PR',
  interesses: [...],
  ativo: true
}

{
  _id: ObjectId('6a3e8b08ab9d566c0e0927df'),
  nome: 'Marina Costa',
  email: 'marina@email.com',
  idade: NumberInt('35'),
  cidade: 'São Paulo',
  estado: 'SP',
  interesses: [...],
  ativo: true,
  premium: true
}
```

ex25:

The screenshot shows the MongoDB Playground interface. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and updating a document in the 'streamflix.usuarios' collection. The right pane displays the result of the update operation.

```
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use Defaults
4 // Make sure you are connected to enable completions and to be able
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with a limit of
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.usuarios.updateOne(
15   { nome: "Carlos Lima" },
16   { $set: { ativo: true } }
17 )
```

Playground Result: streamflix.usuarios

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: NumberInt('1'),
  modifiedCount: NumberInt('1'),
  upsertedCount: NumberInt('0')
}
```

ex26:

The screenshot shows the MongoDB Playground interface in VS Code. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and updating a document in the 'streamflix' database. The right pane shows the result of the update operation.

```
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use Playground
4 // Make sure you are connected to enable completions and to the MongoDB instance
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right pane
7 // By default the first 20 documents will be returned with the sort order
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.updateOne(
15   { titulo: "Cidade de Deus" },
16   { $set: { disponivel: true } }
17 )
```

```
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }
```

ex27:

The screenshot shows the MongoDB Playground interface in VS Code. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and updating a document in the 'streamflix' database. The right pane shows the result of the update operation.

```
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use Playground
4 // Make sure you are connected to enable completions and to the MongoDB instance
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right pane
7 // By default the first 20 documents will be returned with the sort order
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.updateOne(
15   { titulo: "Matrix" },
16   { $set: { idiomaOriginal: "Inglês" } }
17 )
```

```
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }
```

ex28:

The screenshot shows the MongoDB Playground interface in VS Code. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and updating a document in the 'streamflix' database. The right pane shows the result of the update operation.

```
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use Playground
4 // Make sure you are connected to enable completions and to the MongoDB instance
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right pane
7 // By default the first 20 documents will be returned with the sort order
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.updateOne(
15   { titulo: "Interstellar" },
16   { $set: { classificacao: "10+" } }
17 )
```

```
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }
```

ex29:

The screenshot shows the MongoDB Playground interface. The left pane contains the following code:

```
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use I
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to t
6 // The result of the last command run in a playground is sh
7 // By default the first 20 documents will be returned with
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use('streamflix')
14 db.conteudos.updateOne(
15   { titulo: "Avatar" },
16   { $set: { avaliacaoMedia: 9.0 } }
17 )
```

The right pane shows the 'Playground Result' for the 'streamflix.conteudos' collection:

```
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }
```

ex30:

The screenshot shows the MongoDB Playground interface. The left pane contains the following code:

```
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use I
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to t
6 // The result of the last command run in a playground is sh
7 // By default the first 20 documents will be returned with
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use('streamflix')
14 db.conteudos.updateOne(
15   { titulo: "Matrix" },
16   { $inc: { visualizacoes: 1 } }
17 )
```

The right pane shows the 'Playground Result' for the 'streamflix.conteudos' collection:

```
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }
```

ex31:

The screenshot shows the MongoDB Playground interface. The left pane contains the following code:

```
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use I
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to t
6 // The result of the last command run in a playground is sh
7 // By default the first 20 documents will be returned with
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use('streamflix')
14 db.conteudos.updateMany(
15   { disponivel: true },
16   { $inc: { visualizacoes: 1000 } }
17 )
```

The right pane shows the 'Playground Result' for the 'streamflix.conteudos' collection:

```
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('28'),
5   modifiedCount: NumberInt('28'),
6   upsertedCount: NumberInt('0')
7 }
```

ex32:


```

Welcome  JS playground-2.mongodb.js  ×
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to the MongoDB
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right
7 // By default the first 20 documents will be returned with the command
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.updateOne(
15   { titulo: "Matrix" },
16   { $push: { generos: "Clássico" } }
17 )

```

```

Playground Result  Playground Result: streamflix.conteudos  Playground  ...
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }

```

ex33:

```

Welcome  JS playground-2.mongodb.js  ×
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to the MongoDB
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right
7 // By default the first 20 documents will be returned with the command
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.updateOne(
15   { titulo: "Matrix" },
16   { $pull: { generos: "Clássico" } }
17 )

```

```

Playground Result  Playground Result: streamflix.conteudos  Playground  ...
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }

```

ex34:

```

Welcome  JS playground-2.mongodb.js  ×
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to the MongoDB
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right
7 // By default the first 20 documents will be returned with the command
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.usuarios.updateOne(
15   { nome: "Beatriz Nunes" },
16   { $unset: { telefone: "" } }
17 )

```

```

Playground Result  Playground Result: streamflix.conteudos  Playground  ...
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('1'),
5   modifiedCount: NumberInt('1'),
6   upsertedCount: NumberInt('0')
7 }

```

ex35:

The screenshot shows the MongoDB Playground interface. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and performing an updateMany operation on the 'streamflix' database. The right pane shows the 'Playground Result' tab with the following JSON output:

```
1 {
2   acknowledged: true,
3   insertedId: null,
4   matchedCount: NumberInt('2'),
5   modifiedCount: NumberInt('2'),
6   upsertedCount: NumberInt('0')
7 }
```

ex36:

The screenshot shows the MongoDB Playground interface. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and performing a find operation on the 'streamflix' database. The right pane shows the 'Playground Result' tab with the following JSON output:

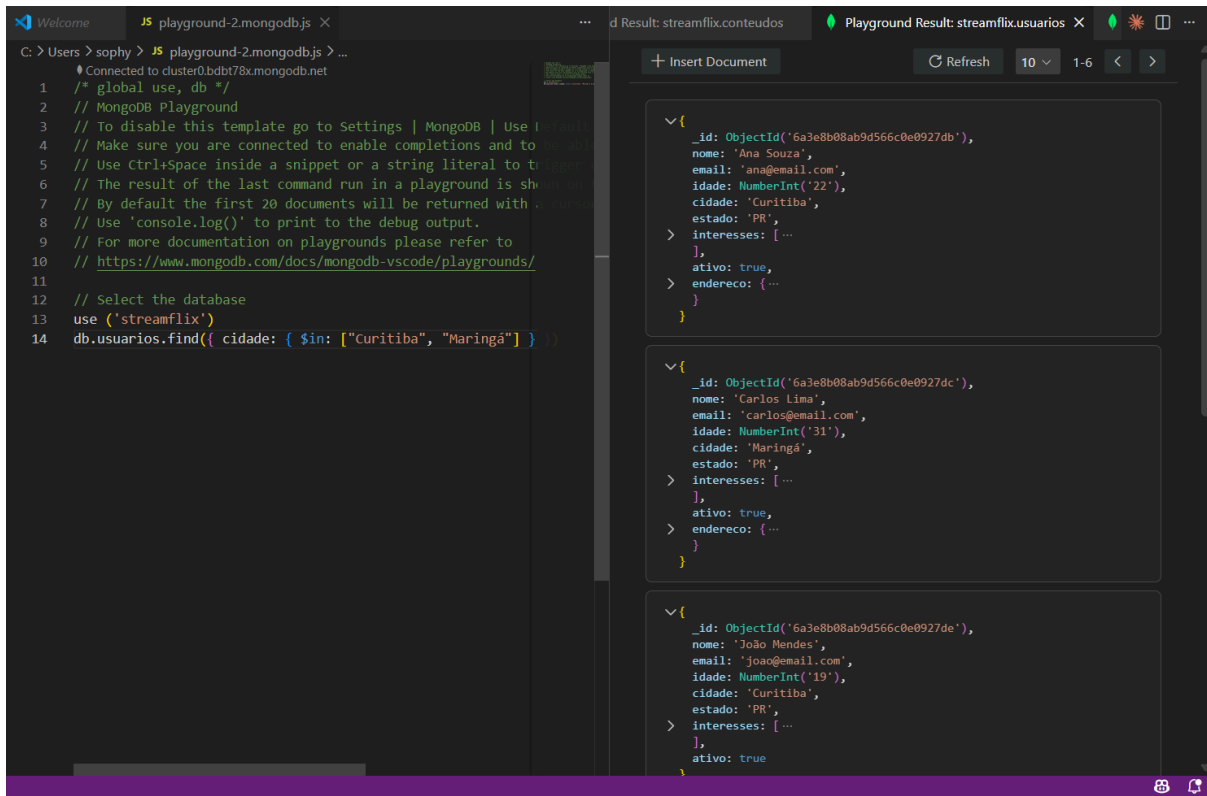
```
+ Insert Document Refresh 10 1-8 < >

1 {
2   _id: ObjectId('6a3e87ce3c1b96d4f3acce49'),
3   titulo: 'Interestelar',
4   tipo: 'filme',
5   ano: NumberInt('2014'),
6   generos: [ ... ],
7   avaliacaoMedia: Double('9.5'),
8   duracaoMinutos: NumberInt('169'),
9   disponivel: true,
10  visualizacoes: NumberInt('2501000'),
11  diretor: { ... },
12  elenco: [ ... ],
13  classificacao: '10+'
14 }

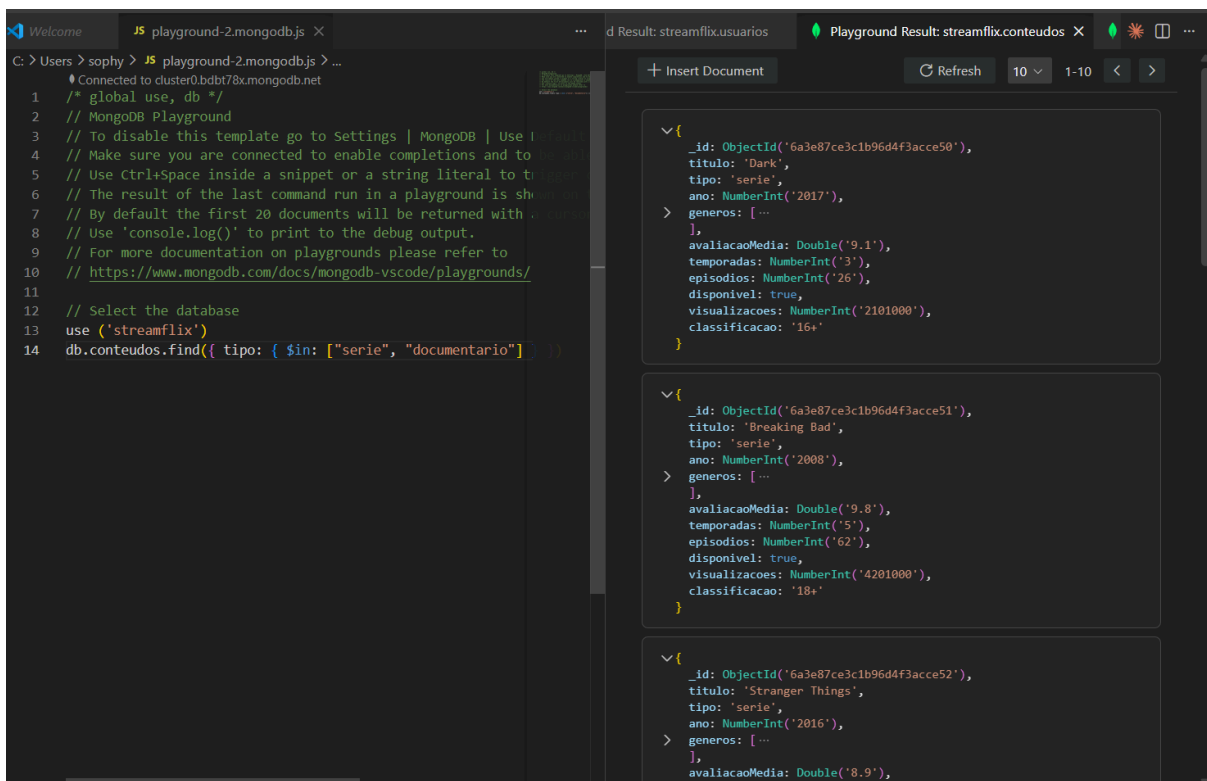
1 {
2   _id: ObjectId('6a3e87ce3c1b96d4f3acce4a'),
3   titulo: 'Matrix',
4   tipo: 'filme',
5   ano: NumberInt('1999'),
6   generos: [ ... ],
7   avaliacaoMedia: Double('9.2'),
8   duracaoMinutos: NumberInt('136'),
9   disponivel: true,
10  visualizacoes: NumberInt('3101001'),
11  diretor: { ... },
12  elenco: [ ... ],
13  idiomaOriginal: 'Inglês'
14 }

1 {
2   _id: ObjectId('6a3e87ce3c1b96d4f3acce4c'),
3   titulo: 'O Senhor dos Anéis: A Sociedade do Anel'
4 }
```

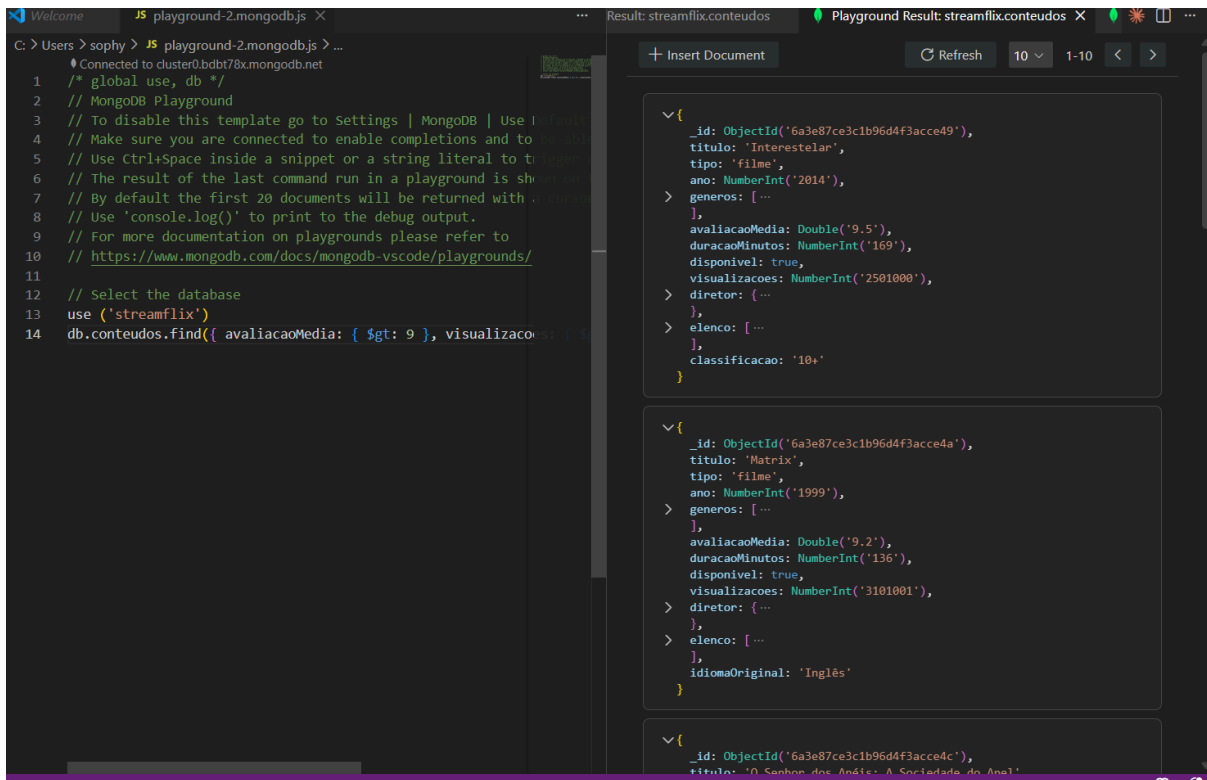
ex37:



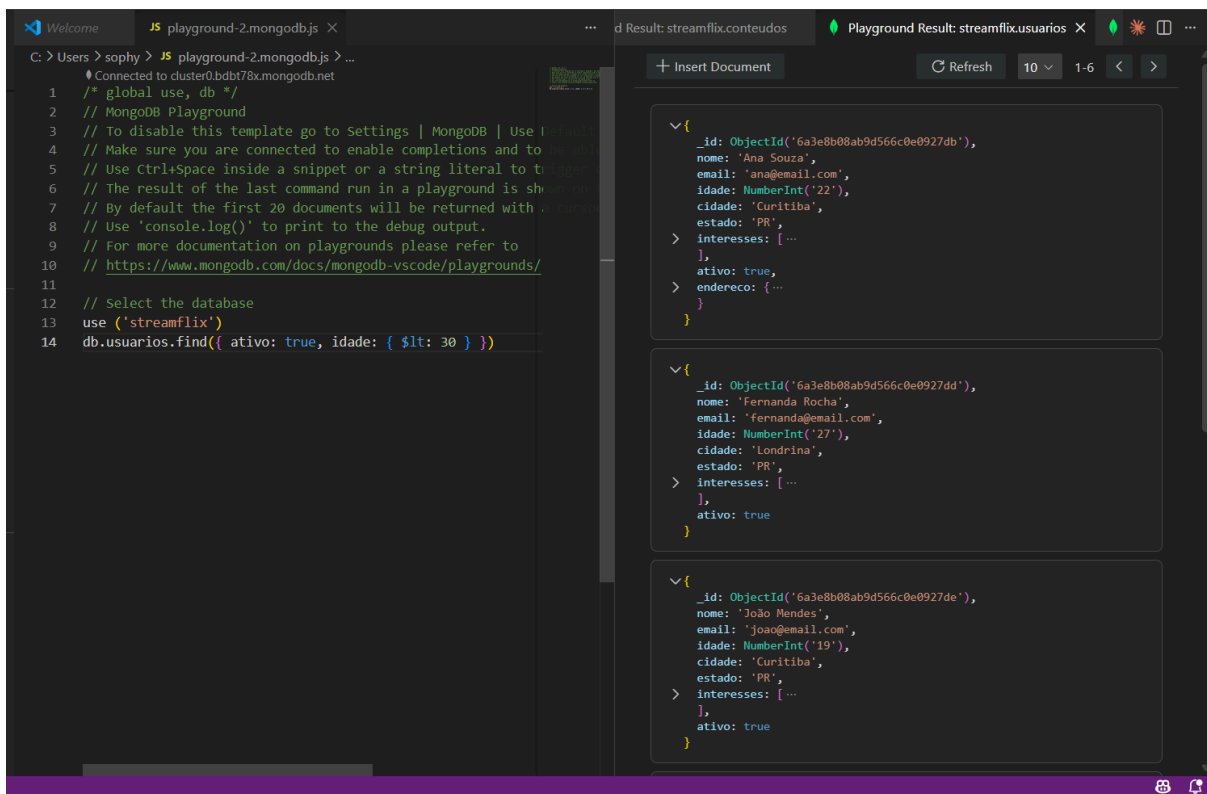
ex38:



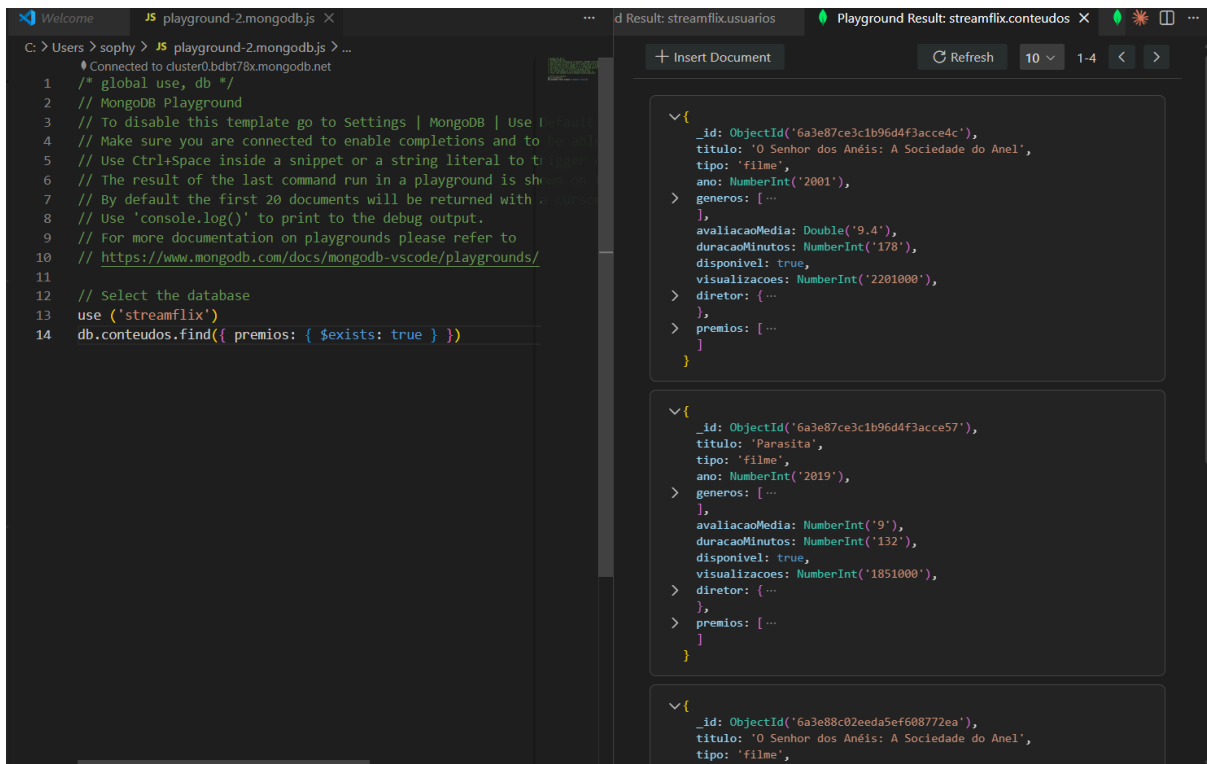
ex39:



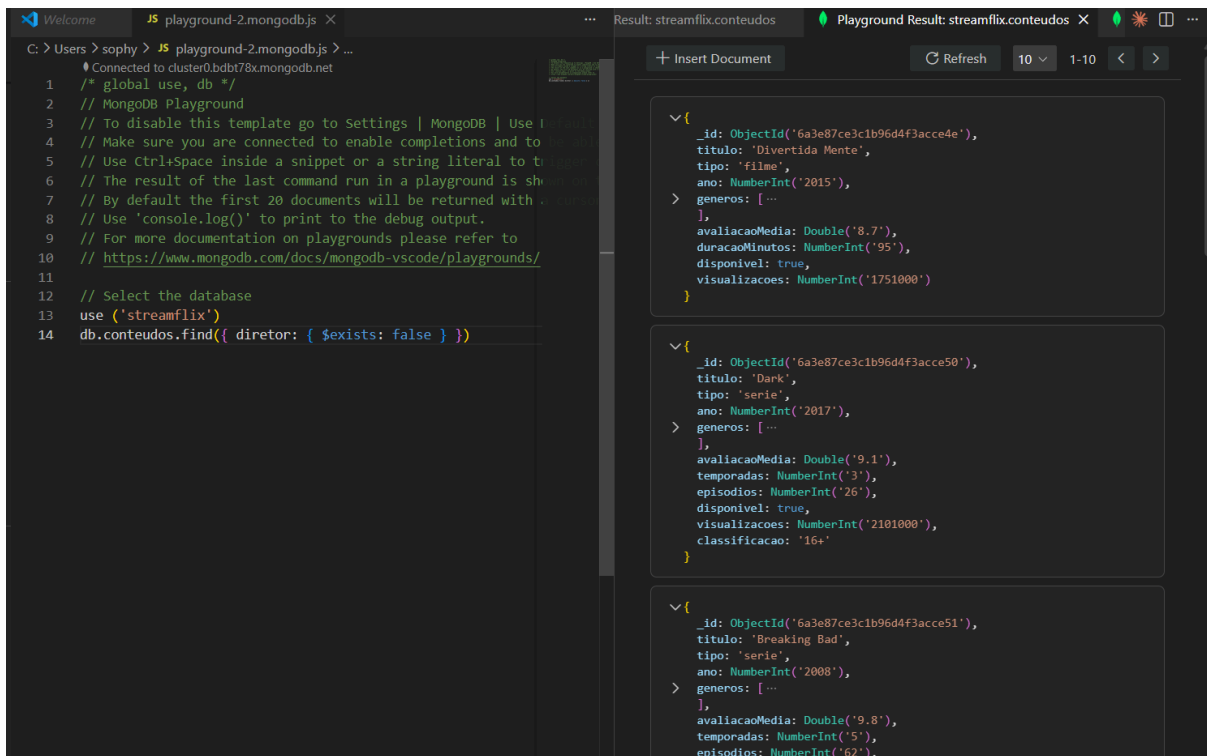
ex40:



ex41:



ex42:



ex43:

The screenshot shows the MongoDB Playground interface. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and querying the 'streamflix' database for users with a premium status. The right pane displays the results of the query, showing two documents with fields like _id, nome, email, idade, cidade, estado, interesses, ativo, and premium.

```
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net

1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.usuarios.find({ premium: { $exists: true } })
```

```
{
  "_id": "6a3e8b08ab9d566c0e0927df",
  "nome": "Marina Costa",
  "email": "marina@email.com",
  "idade": 35,
  "cidade": "São Paulo",
  "estado": "SP",
  "interesses": [...],
  "ativo": true,
  "premium": true
}
```

```
{
  "_id": "6a3e8b08ab9d566c0e0927e4",
  "nome": "Eduardo Martins",
  "email": "eduardo@email.com",
  "idade": 33,
  "cidade": "Curitiba",
  "estado": "PR",
  "interesses": [...],
  "ativo": true,
  "premium": false
}
```

ex44:

The screenshot shows the MongoDB Playground interface. The left pane contains a JavaScript snippet for connecting to a MongoDB cluster and querying the 'streamflix' database for content. The right pane displays the results of the query, showing three documents with fields like _id, titulo, tipo, ano, generos, avaliacaoMedia, temporadas, episodios, disponivel, visualizacoes, and classificacao.

```
C: > Users > sophy > JS playground-2.mongodb.js > ...
  Connected to cluster0.bdbt78x.mongodb.net

1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to trigger
6 // The result of the last command run in a playground is shown on
7 // By default the first 20 documents will be returned with
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.find({ temporadas: { $exists: true } })
```

```
{
  "_id": "6a3e87ce3c1b96d4f3acce50",
  "titulo": "Dark",
  "tipo": "serie",
  "ano": 2017,
  "generos": [...],
  "avaliacaoMedia": 9.1,
  "temporadas": 3,
  "episodios": 26,
  "disponivel": true,
  "visualizacoes": 2101000,
  "classificacao": "16+"
}
```

```
{
  "_id": "6a3e87ce3c1b96d4f3acce51",
  "titulo": "Breaking Bad",
  "tipo": "serie",
  "ano": 2008,
  "generos": [...],
  "avaliacaoMedia": 9.8,
  "temporadas": 5,
  "episodios": 62,
  "disponivel": true,
  "visualizacoes": 4201000,
  "classificacao": "18+"
}
```

```
{
  "_id": "6a3e87ce3c1b96d4f3acce52",
  "titulo": "Stranger Things",
  "tipo": "serie",
  "ano": 2016,
  "generos": [...],
  "avaliacaoMedia": 8.9,
  "temporadas": 4
}
```

ex45:

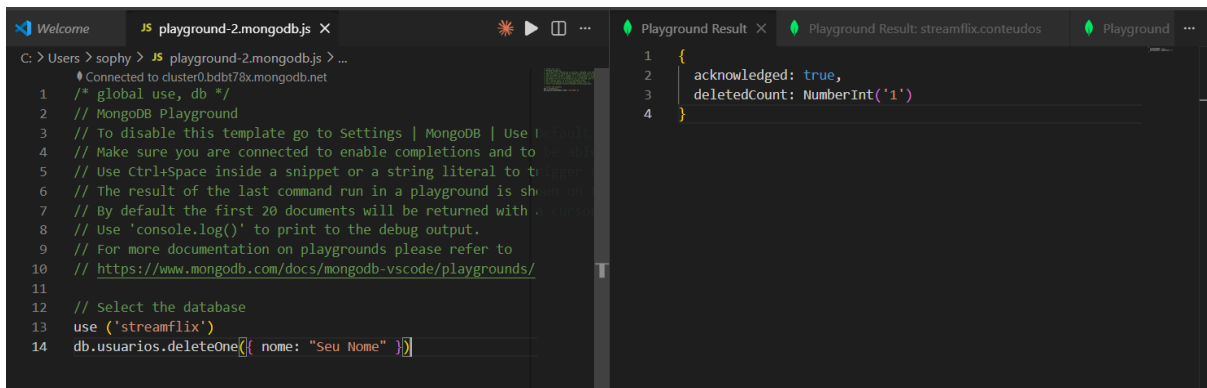
Exercício 45 — explicação:

O MongoDB é um banco NoSQL orientado a documentos, e por isso não exige que todos os documentos de uma coleção tenham o mesmo schema. Cada documento é independente e pode ter sua própria estrutura de campos.

Na coleção **conteudos**, isso faz sentido porque ela armazena tipos diferentes (filmes, séries, documentários), e cada tipo tem características próprias. Uma série tem

temporadas, um filme não precisa disso. Nem todo conteúdo possui **premios**. Essa flexibilidade evita campos vazios desnecessários e permite que a estrutura dos dados evolua sem precisar de migrações, como seria exigido em um banco relacional.

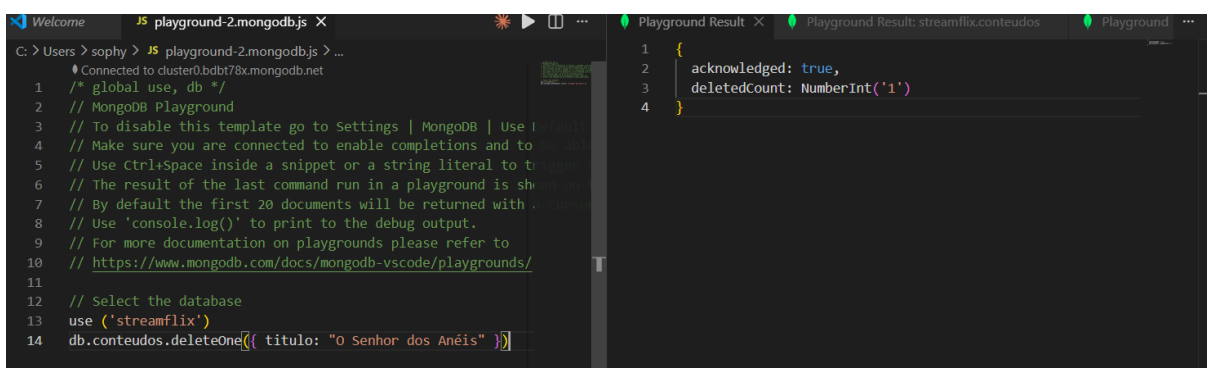
ex46:



```
C: > Users > sophy > JS playground-2.mongodb.js > ...
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB Playground
4 // Make sure you are connected to enable completions and to the MongoDB instance
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right side
7 // By default the first 20 documents will be returned with the command db.collection.find()
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.usuarios.deleteOne({ nome: "Seu Nome" })
```

```
1 {
2   acknowledged: true,
3   deletedCount: NumberInt('1')
4 }
```

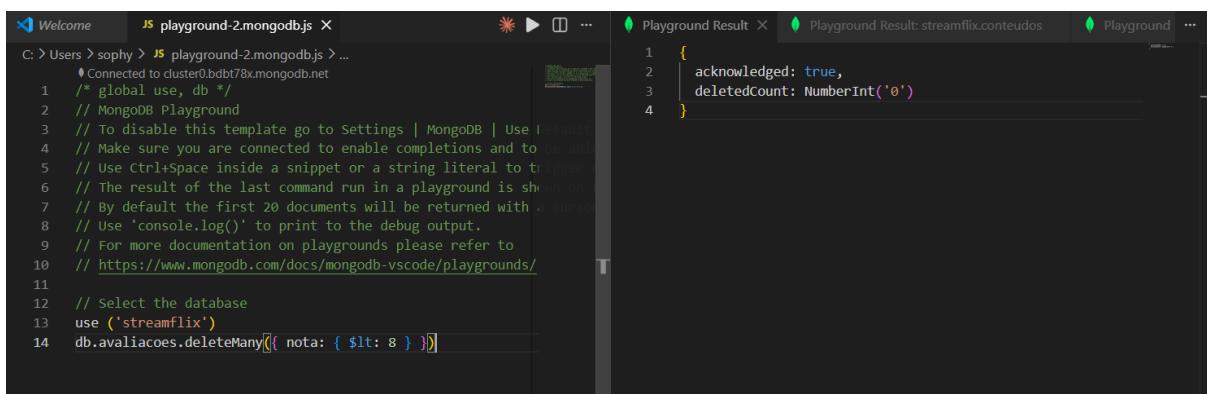
ex47:



```
C: > Users > sophy > JS playground-2.mongodb.js > ...
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB Playground
4 // Make sure you are connected to enable completions and to the MongoDB instance
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right side
7 // By default the first 20 documents will be returned with the command db.collection.find()
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.conteudos.deleteOne({ titulo: "O Senhor dos Anéis" })
```

```
1 {
2   acknowledged: true,
3   deletedCount: NumberInt('1')
4 }
```

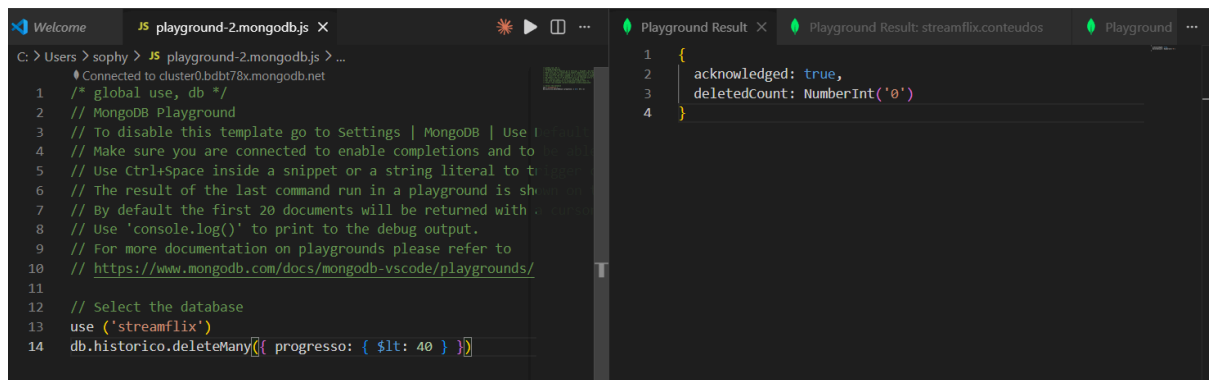
ex48:



```
C: > Users > sophy > JS playground-2.mongodb.js > ...
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use MongoDB Playground
4 // Make sure you are connected to enable completions and to the MongoDB instance
5 // Use Ctrl+Space inside a snippet or a string literal to trigger completions
6 // The result of the last command run in a playground is shown on the right side
7 // By default the first 20 documents will be returned with the command db.collection.find()
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.avaliacoes.deleteMany({ nota: { $lt: 8 } })
```

```
1 {
2   acknowledged: true,
3   deletedCount: NumberInt('0')
4 }
```

ex49:



```
C: > Users > sophy > JS playground-2.mongodb.js > ...
Connected to cluster0.mongodb.net
1 /* global use, db */
2 // MongoDB Playground
3 // To disable this template go to Settings | MongoDB | Use I
4 // Make sure you are connected to enable completions and to
5 // Use Ctrl+Space inside a snippet or a string literal to t
6 // The result of the last command run in a playground is sh
7 // By default the first 20 documents will be returned with .
8 // Use 'console.log()' to print to the debug output.
9 // For more documentation on playgrounds please refer to
10 // https://www.mongodb.com/docs/mongodb-vscode/playgrounds/
11
12 // Select the database
13 use ('streamflix')
14 db.historico.deleteMany({ progresso: { $lt: 40 } })
```

```
1 {
2   acknowledged: true,
3   deletedCount: NumberInt('0')
4 }
```

ex50:

Manter as informações em várias coleções (referência) significa dividir os dados relacionados em documentos distintos, ligados por um identificador, como o `_id` de um usuário guardado dentro de um documento de assinatura. É parecido com o modelo relacional, exige que você faça buscas separadas ou use `$lookup` para juntar os dados, mas evita duplicação e mantém os documentos menores.

Já armazenar tudo em um único documento significa embutir as informações relacionadas diretamente dentro do documento principal, como os dados de endereço dentro do próprio documento de usuário. Isso facilita a leitura, já que tudo vem em uma única consulta, mas pode gerar duplicação de dados e documentos muito grandes se a relação envolver muitos registros.

ex51:

Vantagem: a leitura é mais rápida e simples, pois todos os dados relacionados vêm em uma única consulta, sem precisar de joins ou buscas adicionais.

Desvantagem: se a informação aninhada se repete em vários lugares, ela pode causar duplicação de dados e dificultar atualizações, já que é preciso alterar a mesma informação em vários documentos.

ex52:

Referência é melhor quando os dados se repetem entre muitos documentos, quando a relação envolve um volume grande de registros que cresce com o tempo (como o histórico de visualizações de um usuário), ou quando os dados relacionados são atualizados com frequência e de forma independente entre si.

ex53:

Dados incorporados são melhores quando a informação é pequena, fixa e sempre consultada junto com o documento principal, como o endereço de um usuário. Também é indicado quando não há necessidade de acessar aquela informação separadamente, e quando ela não é compartilhada entre outros documentos.